University of Glasgow

James Watt School of Engineering

ENG4184

Navigation Systems 4

IMPORTANT STUDENT INSTRUCTIONS
Ensure your **Student Number** and **date of birth** are **entered** in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers) 2 8 3 9 7 6 8
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Date of Birth (DD/MM/YYYY) 0 9 / 0 3 / 2 0 0 4
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COMPULSORY Q2.
Do not mark any other boxes.

☐ DO NOT USE ☒ **Q2** ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE

You may use both sides but only answer **ONE** question per sheet

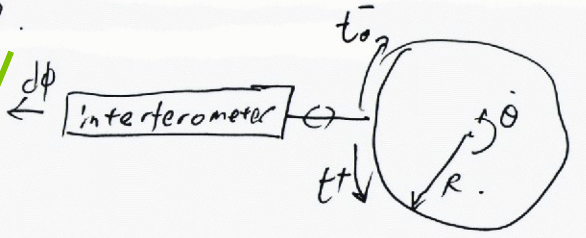
Q2 (a).

[1] (i) An inertially stabilised platform uses inertial sensors mounted on inertially stabilised platform placed on a 3-axis gimbal. The sensor's measured signal can be used as feedback signal for the torque morter on each axis of the gimbal to keep stabilised. The attitude can be obtained according to the measurment.

[1] (ii) * mechanically complex
* costly for manufacture / maintenance.
* larger volumn / weight
* cannot be directly used for airframe navigation.

[2] (iii) Strapdown mounts the accelerometers and gyros rigidly on the vehicle. It can measure the specific force and angular velocity on the body's frame and use them to calculate the attitude.
It's a better choice but computationally complex.

[12] (b).



$$t^- = \frac{2\pi R}{C + \dot\theta R} = \frac{\frac{2\pi R}{C}}{1 + \frac{R}{C}\dot\theta} \approx \frac{2\pi R}{C}\left(1 - \frac{R}{C}\dot\theta\right)$$

$$t^+ = \frac{2\pi R}{C - \dot\theta R} = \frac{\frac{2\pi R}{C}}{1 - \frac{R}{C}\dot\theta} \approx \frac{2\pi R}{C}\left(1 + \frac{R}{C}\dot\theta\right)$$

$$\Rightarrow \Delta t = t^+ - t^- = \frac{2\pi R}{C} \cdot 2 \cdot \frac{R}{C}\dot\theta = \frac{4\pi R^2}{C^2}\dot\theta$$

Therefore, the $\Delta L = C\Delta t = \frac{4\pi R^2}{C}\dot\theta$.

From $\Delta L = \frac{d\phi}{2\pi}\cdot\lambda$

$$\Rightarrow \frac{4\pi R^2}{C}\dot\theta = \frac{\lambda}{2\pi}\,d\phi$$

$$\Rightarrow \dot\theta = \frac{\lambda c}{8\pi^2 R^2}\,d\phi$$

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Q2 Continued

[4]

(c) Bias error : the constant offset of the measurement.

Scaling factor : the error due to the gradient between the input and output.

Cross-coupling error : The misalignment between the senstive axis of sensor and the orthogonal axis of the body frame.

Random noise : The thermo-mechanical noise.